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(54) **NON-VOLATILE CURRENT MIRROR
CIRCUIT WITH PROGRAMMABLE
TRANSISTOR**

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(52) **U.S. Cl.**
CPC **G05F 3/262** (2013.01)

(58) **Field of Classification Search**
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See application file for complete search history.

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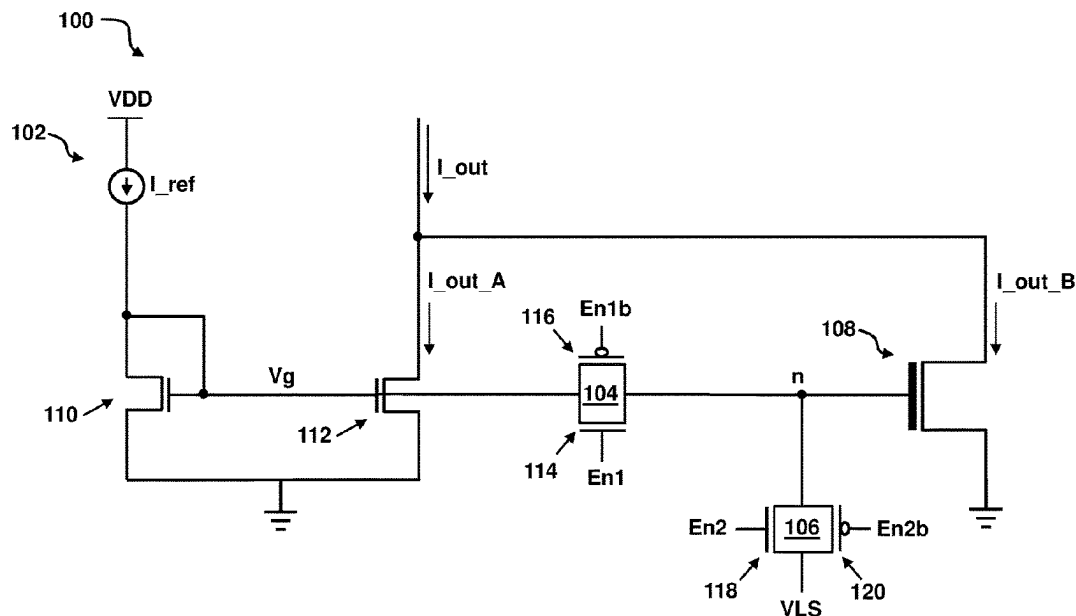
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(57) **ABSTRACT**

Embodiments of the present disclosure provide a structure,
including: a current mirror; a programmable transistor with
an adjustable threshold voltage connected in parallel with an
output transistor of the current mirror; an enable switch for
coupling the current mirror to a gate of the programmable
transistor; and a programming switch coupled to the gate of
the programmable transistor.

17 Claims, 11 Drawing Sheets



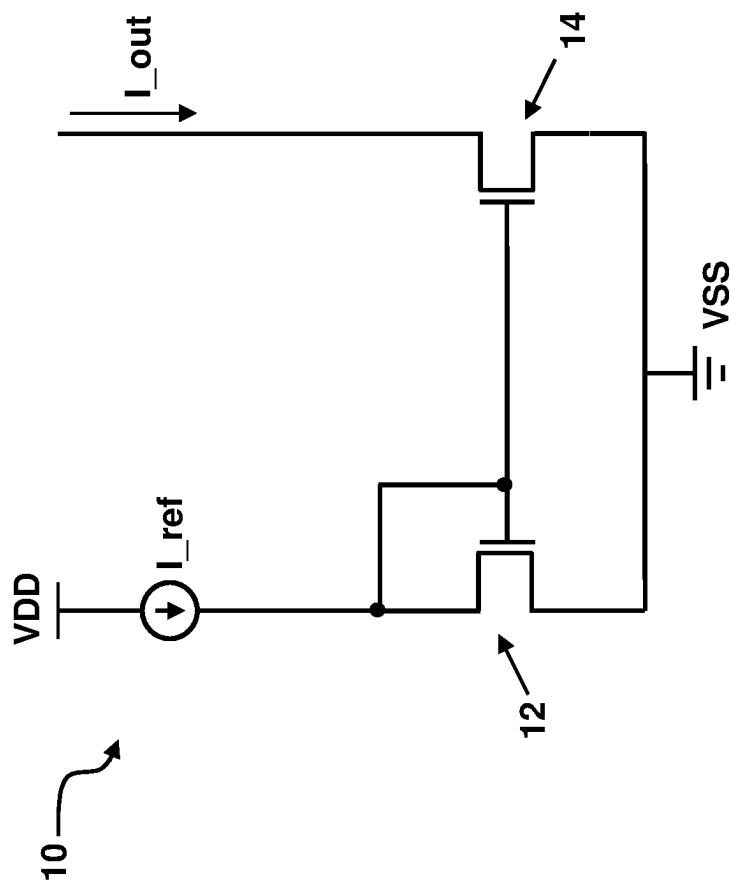


FIG. 1

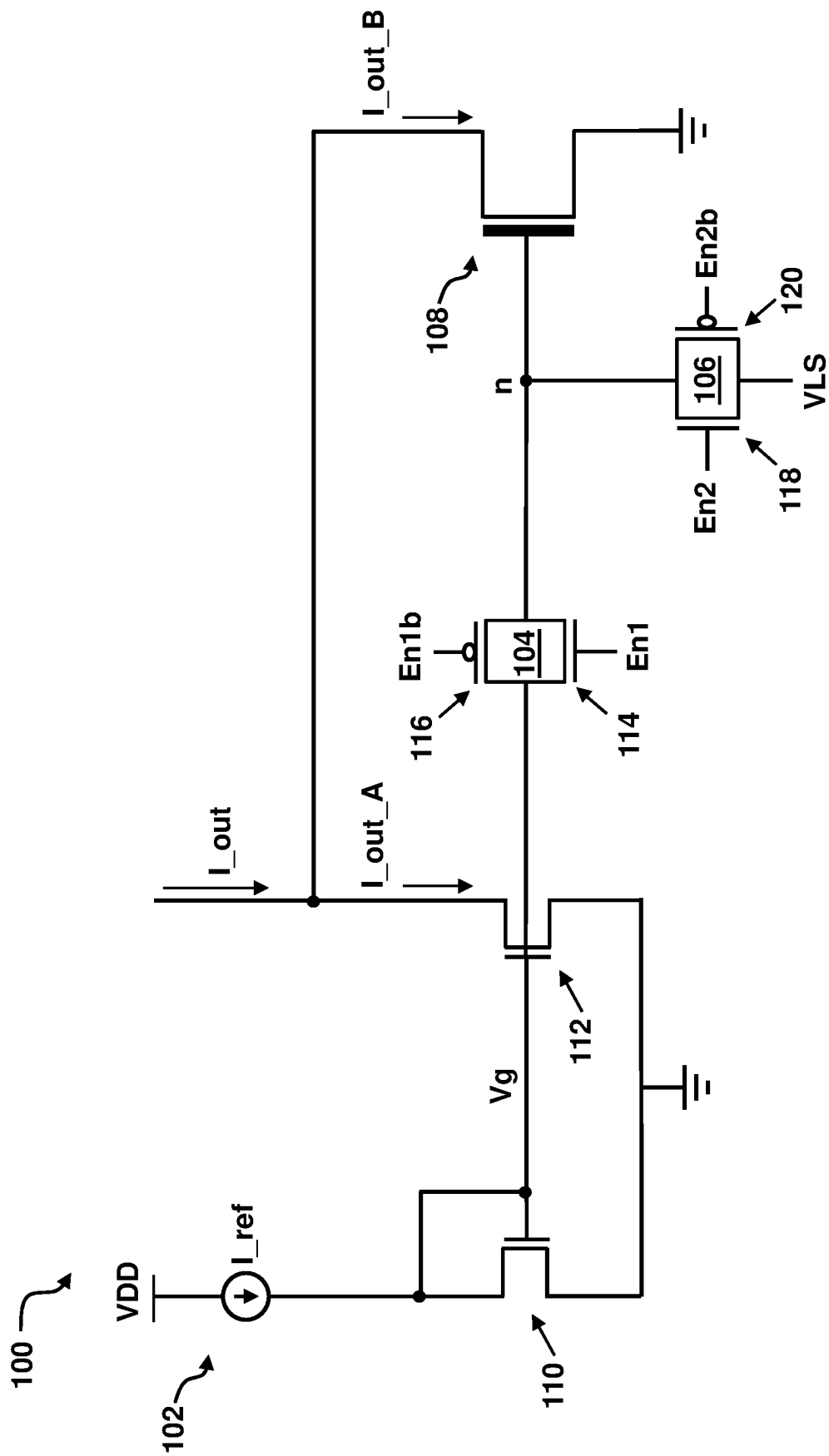


FIG. 2

N-type FeFET, Low VT Programming

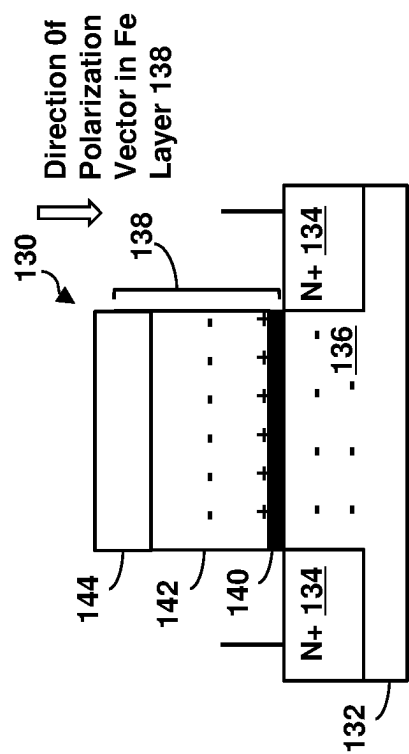


FIG. 3A

N-type FeFET, High VT Programming

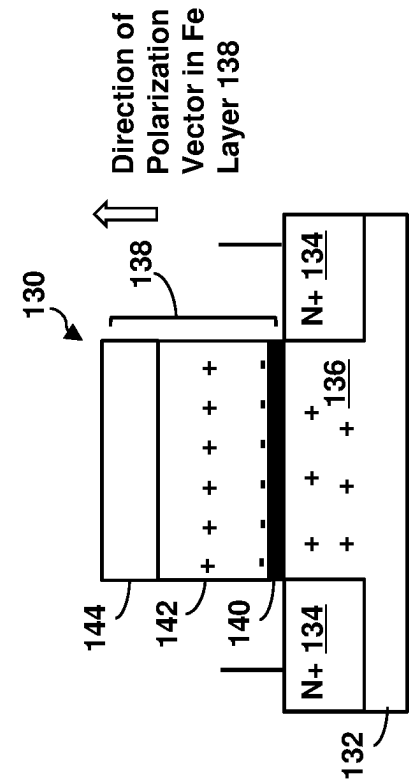


FIG. 3B

N-type CTFET, High VT Programming

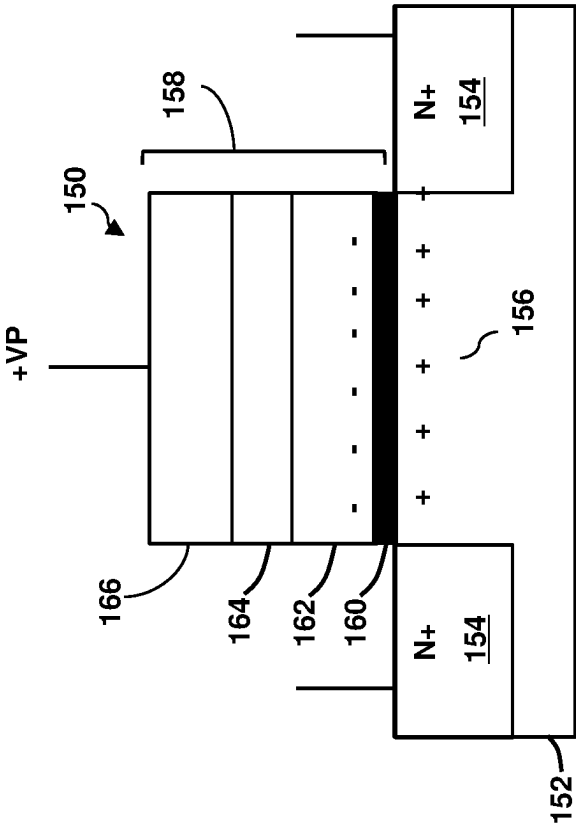


FIG. 4B

N-type CTFET, Low VT Programming

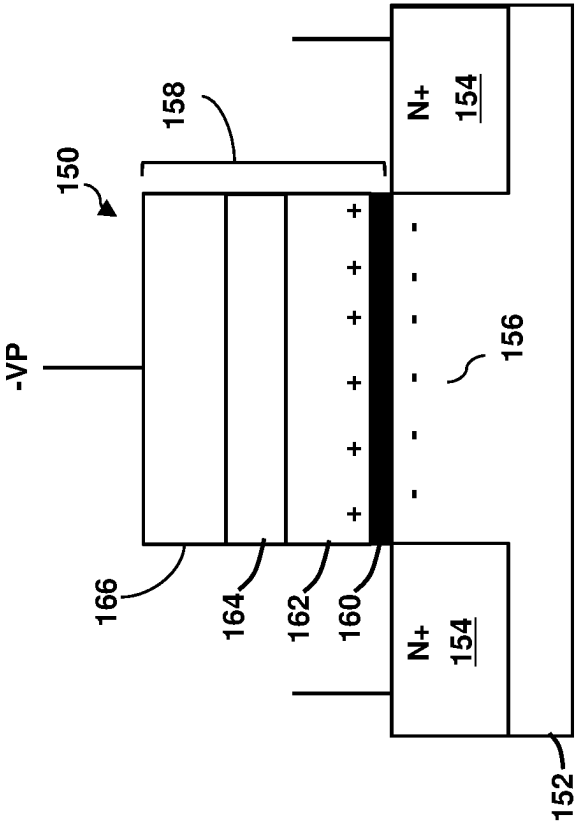
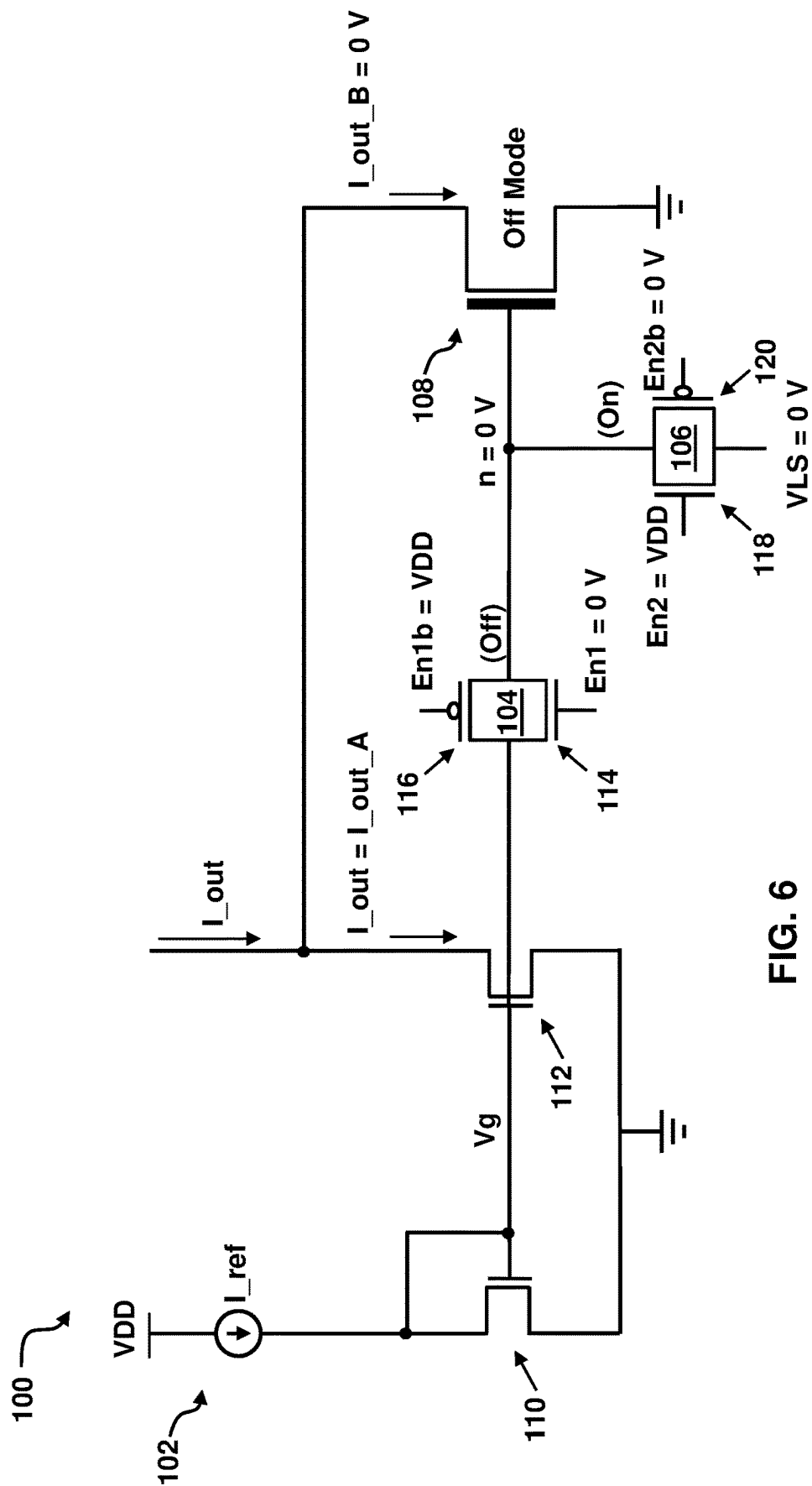


FIG. 4A

VLS	En1	En1b	En2	En2b	Vg	n	Enab. Switch	Prog. Switch	Mode
0	0	VDD	VDD	0	< VDD	0	OFF	ON	Off
0	VDD	0	0	VDD	< VDD	Vg	ON	OFF	Current Mirror
VP	0	VP	VP	0	< VDD	VP	OFF	ON	Program
-Ve	-Ve- VTH	VDD	0	-Ve	< VDD	-Ve	OFF	ON	Erase

FIG. 5



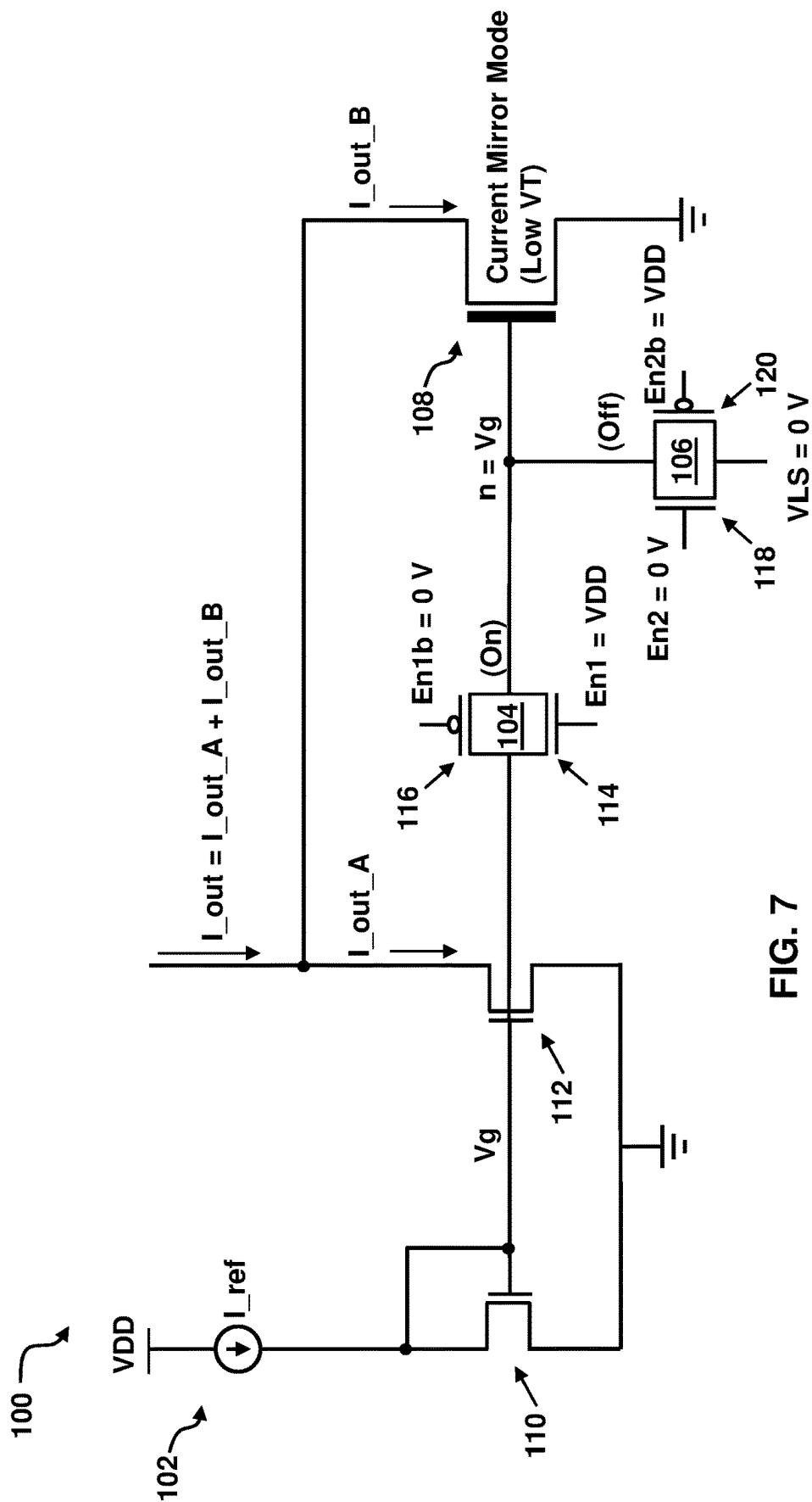
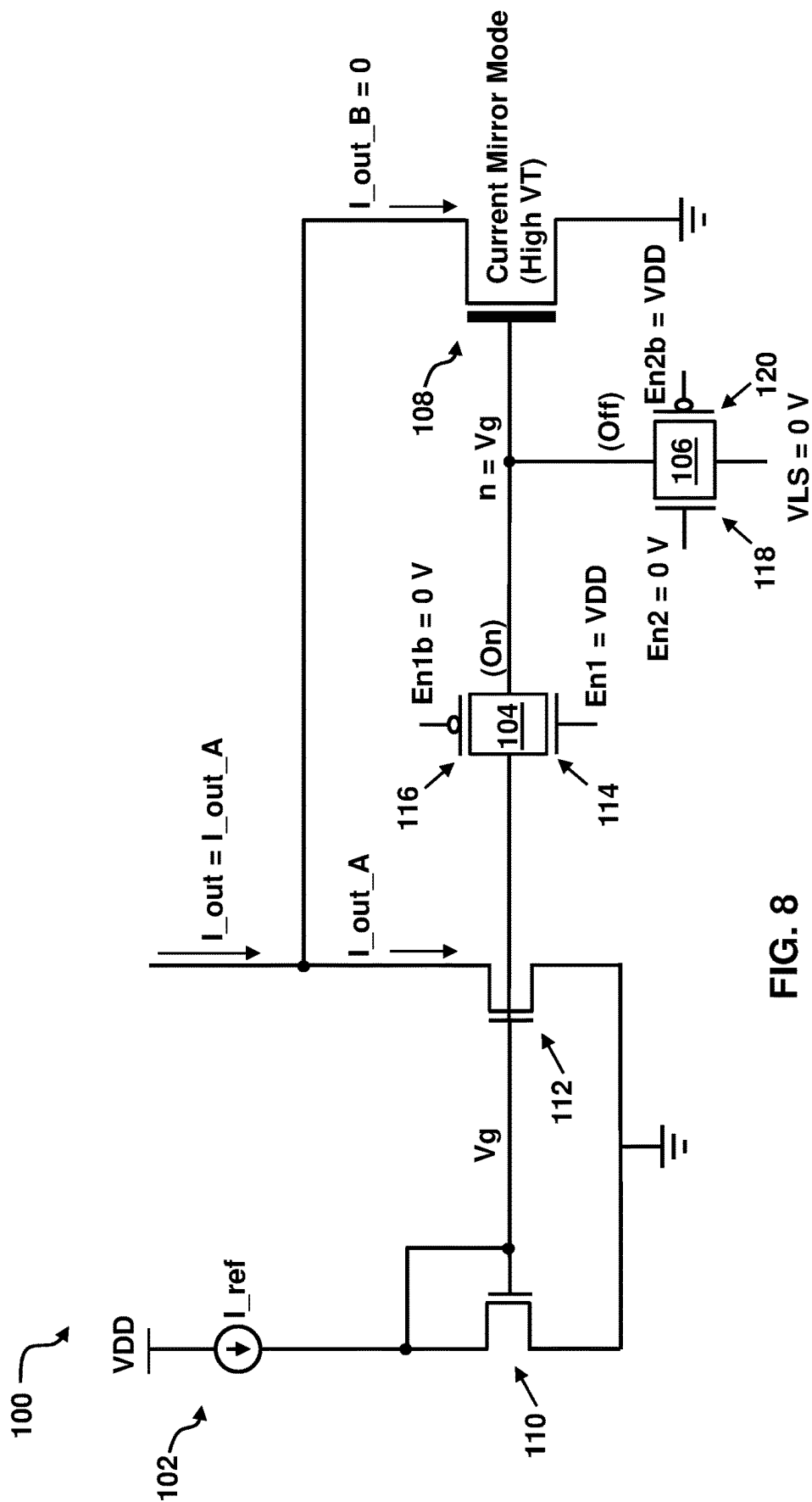


FIG. 7



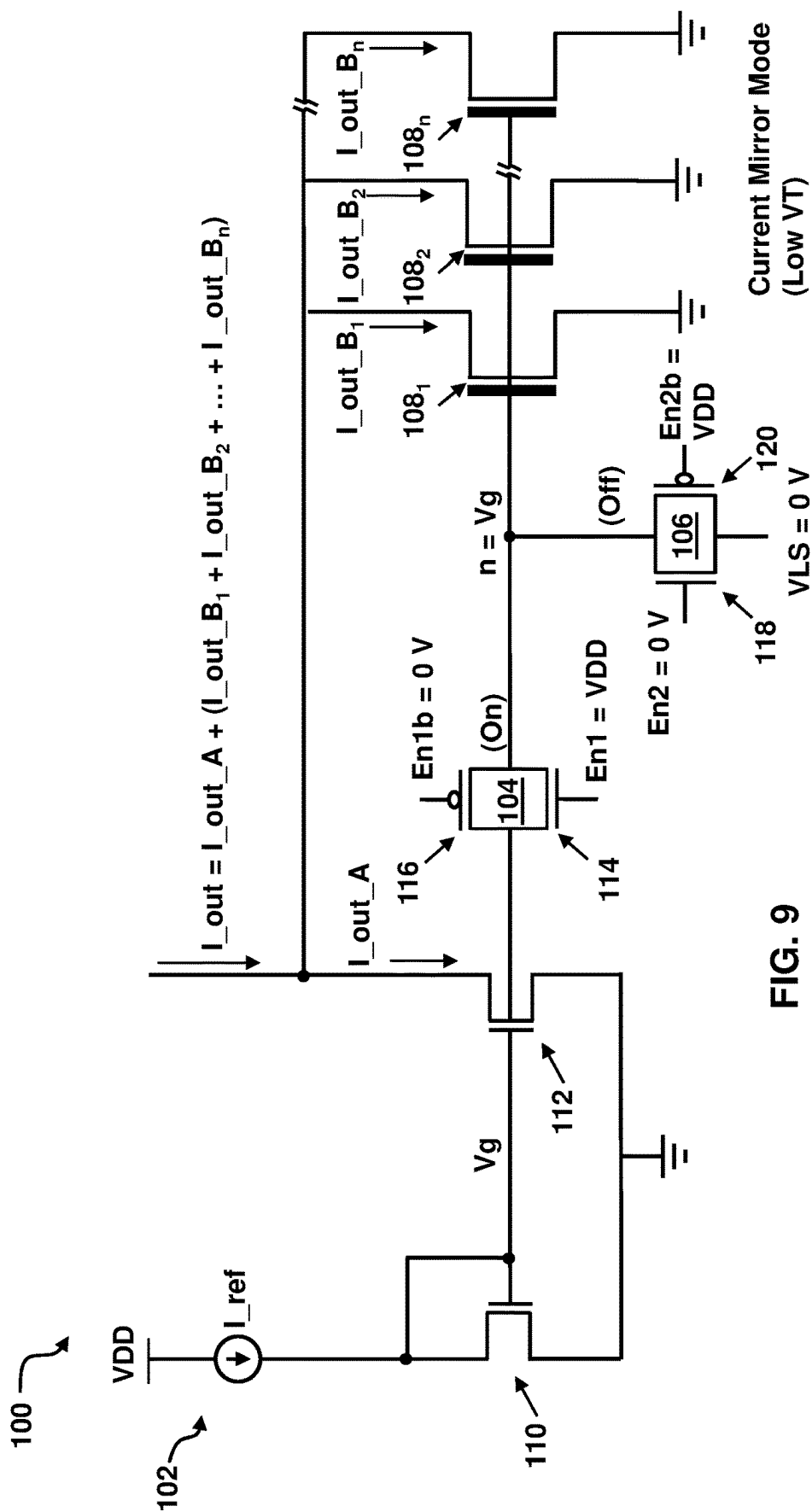


FIG. 9

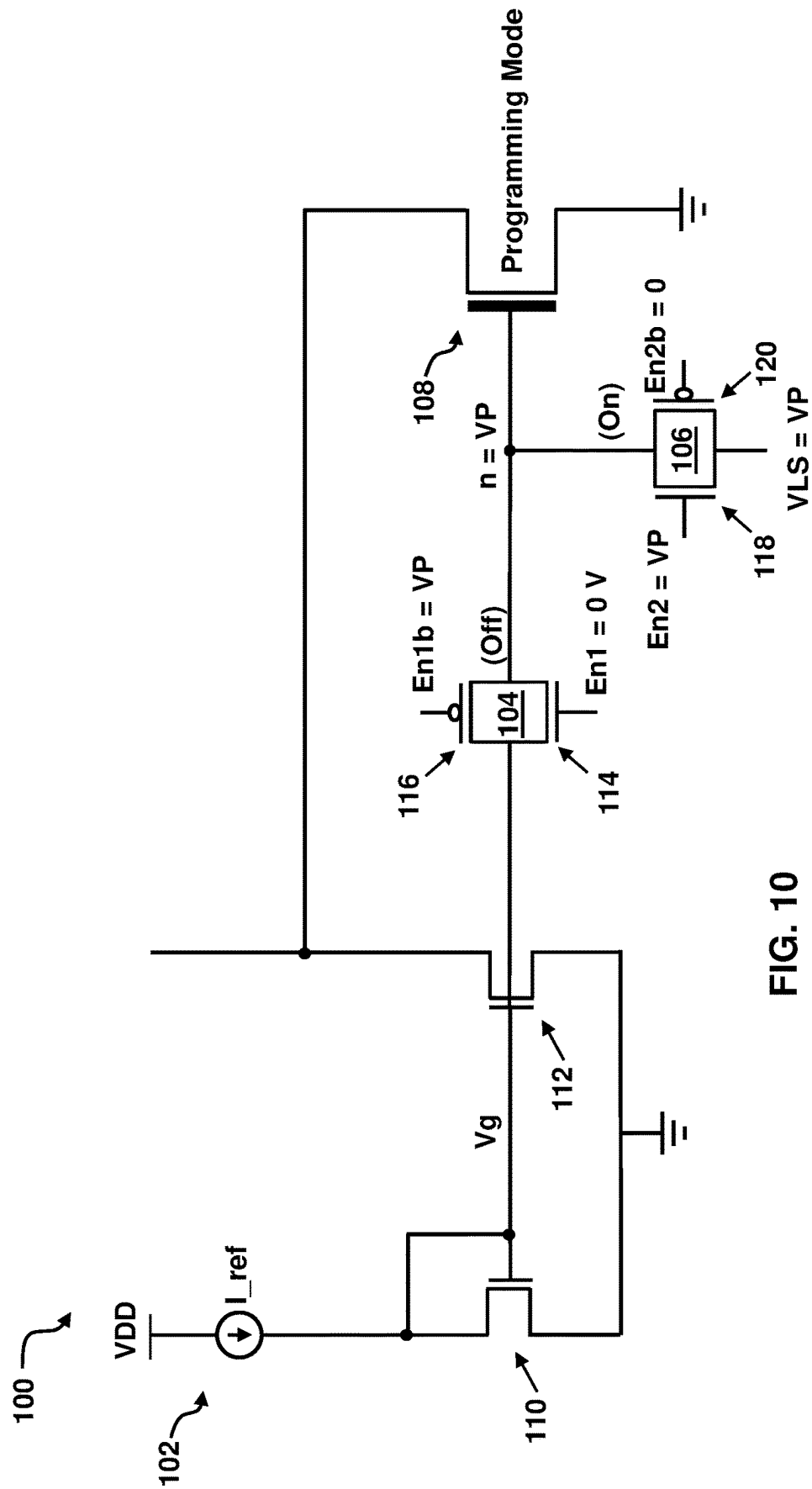


FIG. 10

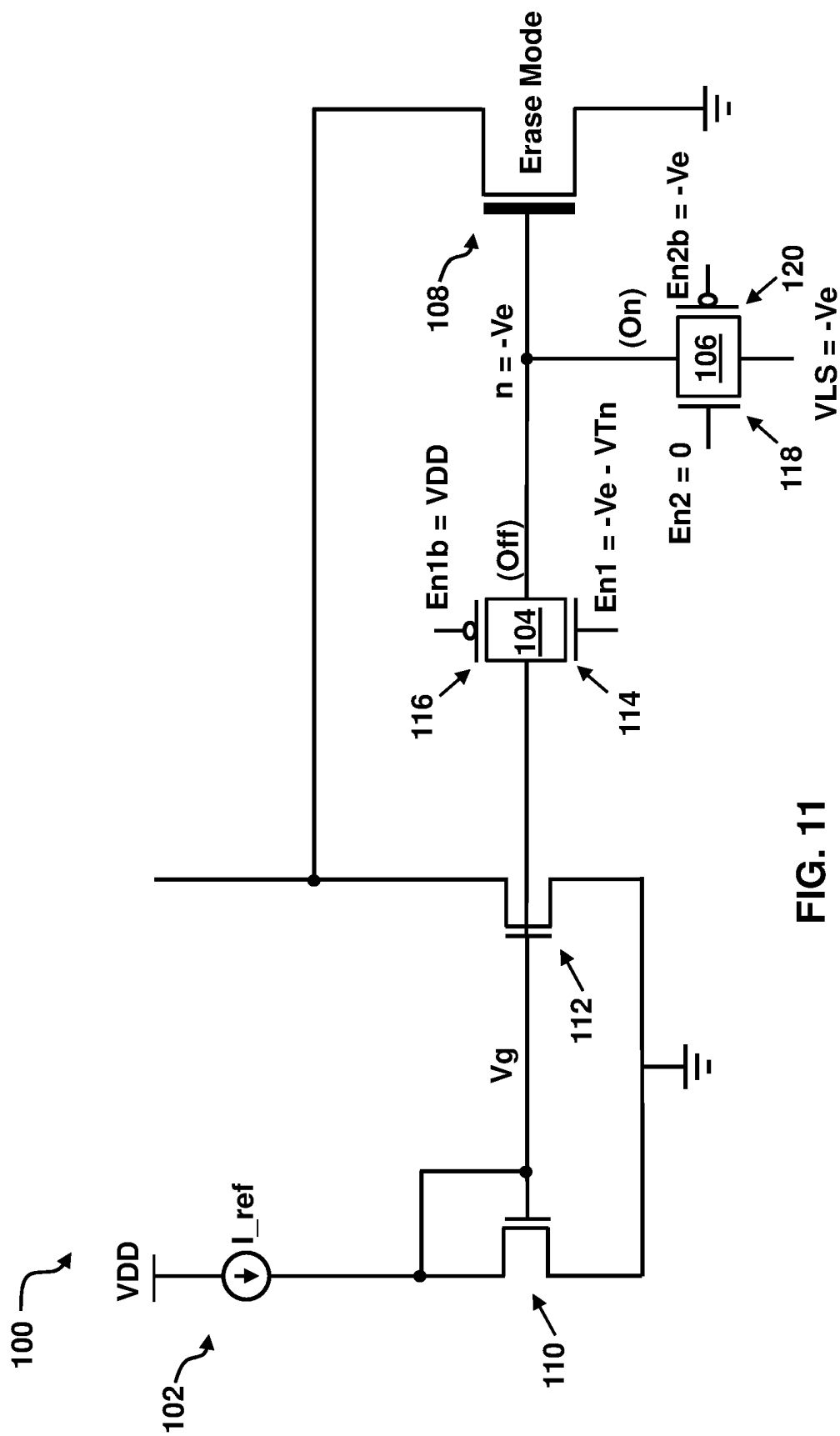


FIG. 11

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NON-VOLATILE CURRENT MIRROR CIRCUIT WITH PROGRAMMABLE TRANSISTOR

BACKGROUND

Embodiments of the disclosure relate generally to current mirrors. More specifically, the disclosure provides a non-volatile current mirror circuit with a programmable transistor.

Current mirrors are often used in analog circuits to generate an output current proportional to an input current. The proportionality constant of the input and output currents in a current mirror is commonly known as the mirroring ratio and is dependent, for example, on the size ratio of the input and output transistors used. Generally, a current mirror **10** can be realized using a pair of field effect transistors (FETs) as depicted in FIG. **1**.

As shown in FIG. **1**, the current mirror **10** includes an N-type reference FET **12**, which has its drain and gate coupled together, and an N-type output FET **14**, which has its gate coupled to the gate of the FET **12**. When a reference current I_{ref} flows through the FET **12**, the voltage at the gate of the FET **12** controls the output current I_{out} flowing through the output FET **14**. As a result, the reference current I_{ref} is mirrored (or proportionally mirrored) by the output current I_{out} .

The size ratio of the FETs **12**, **14**, and hence the mirroring ratio, of the current mirror **10** may be altered by connecting a plurality of the output FETs **14** in parallel. By adding switches controlled, for example, by externally stored binary control bits, in series with the parallel-connected output FETs **14**, the number of output FETs **14** active in the current mirror **10** at any given moment can be changed by controlling the switches, and in this manner the mirroring ratio can be dynamically controlled (e.g., programmed).

SUMMARY

Aspects of the disclosure provide a structure, including: a current mirror; a programmable transistor with an adjustable threshold voltage connected in parallel with an output transistor of the current mirror; an enable switch for coupling the current mirror to a gate of the programmable transistor; and a programming switch coupled to the gate of the programmable transistor.

Another aspect of the disclosure includes a current mirror circuit, including a current mirror; a programmable transistor with an adjustable threshold voltage connected in parallel with an output transistor of the current mirror, wherein the programmable transistor is operable in any of an off mode, a current mirror mode, a programming mode, and an erase mode; an enable switch for coupling the current mirror to a gate of the programmable transistor; and a programming switch coupled to the gate of the programmable transistor for controlling the adjustable threshold voltage of the programmable transistor.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** depicts a conventional current mirror.

FIG. **2** depicts a current mirror circuit with a programmable transistor according to embodiments of the disclosure.

FIGS. **3A** and **3B** are cross-sectional diagrams illustrating different threshold voltage (VT) states of an N-type ferroelectric field effect transistor (FeFET) that can be incorpo-

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rated into the current mirror circuit of FIG. **2** as a programmable transistor according to embodiments of the disclosure.

FIGS. **4A** and **4B** are cross-sectional diagrams illustrating different VT states of an N-type charge trap field effect transistor (CTFET) that can be incorporated into the current mirror circuit of FIG. **2** as a programmable transistor according to embodiments of the disclosure.

FIG. **5** is a table illustrating signals for controlling the operation of the current mirror circuit of FIG. **2** according to embodiments of the disclosure.

FIG. **6** depicts the current mirror circuit of FIG. **2** with the programmable transistor in an off mode according to embodiments of the disclosure.

FIG. **7** depicts the current mirror circuit of FIG. **2** with the programmable transistor in a current mirror mode and with a low VT according to embodiments of the disclosure.

FIG. **8** depicts the current mirror circuit of FIG. **2** with the programmable transistor in a current mirror mode and with a high VT according to embodiments of the disclosure.

FIG. **9** depicts a current mirror circuit including a plurality of programmable transistors according to embodiments of the disclosure.

FIG. **10** depicts the current mirror circuit of FIG. **2** with the programmable transistor in a programming mode according to embodiments of the disclosure.

FIG. **11** depicts the current mirror circuit of FIG. **2** with the programmable transistor in an erase mode according to embodiments of the disclosure.

It is noted that the drawings of the disclosure are not necessarily to scale. The drawings are intended to depict only typical aspects of the disclosure, and therefore should not be considered as limiting the scope of the disclosure. In the drawings, like numbering represents like elements between the drawings.

DETAILED DESCRIPTION

In the following description, reference is made to the accompanying drawings that form a part thereof, and in which is shown by way of illustration specific exemplary embodiments in which the present teachings may be practiced. These embodiments are described in sufficient detail to enable those skilled in the art to practice the present teachings, and it is to be understood that other embodiments may be used and that changes may be made without departing from the scope of the present teachings. The following description is, therefore, merely illustrative.

FIG. **2** depicts a non-volatile programmable current mirror circuit **100** according to embodiments of the disclosure. In general, the current mirror circuit **100** includes a current mirror **102**, an enable switch **104**, a programming switch **106**, and a programmable transistor **108** with a threshold voltage (VT) that is programmable (e.g., either low or high) in order to selectively adjust the total output current I_{out} of the current mirror circuit **100** and to do so without relying on externally stored binary control bits.

Similar to the current mirror **10** depicted in FIG. **1**, the current mirror **102** in FIG. **2** includes an N-type reference field effect transistor (FET) **110** and an N-type output FET **112**. As shown, the drain and the gate of the FET **110** are tied together. The gate of the FET **110** is also connected to the gate of the output FET **112**. Ignoring for the moment the effect of the programmable transistor **108** on the output current I_{out} , when a reference current I_{ref} flows through the FET **110**, the voltage V_g at the gate of the FET **110** controls the output current I_{out} flowing through the output

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FET **112**. To this extent, the reference current I_{ref} is mirrored (or proportionally mirrored) by the output current I_{out} .

The enable switch **104** can be a complementary metal oxide semiconductor (CMOS)-based switch (also referred to herein as a transmission gate). Such a CMOS-based switch can include an N-type FET (NFET) **114** coupled in parallel to a P-type FET (PFET) **116**, with the drain and source of the NFET **114** tied to the drain and source, respectively, of the PFET **116**. A first enable signal $En1$ is provided to the gate of the NFET **114** and an inverse of the first enable signal $En1b$ is provided to the gate of the PFET **116**. In general, when the first enable signal $En1$ is high and the inverse of the first enable signal $En1b$ is low, the enable switch **104** is "ON" and the voltage V_g at the input of the enable switch **104** is passed to an output (node n) of the enable switch **104**. When the first enable signal $En1$ is low and the inverse of the first enable signal $En1b$ is high, the enable switch **104** is "OFF" and the voltage V_g at the input of the enable switch **104** is not passed to node n . Alternatively, the enable switch **104** could be any other type of switch suitable for selectively passing V_g to node n when "ON" or preventing passage of V_g to node n when "OFF". As shown in FIG. 2, node n is coupled to the gate of the programmable transistor **108**.

Similar to the enable switch **104**, the programming switch **106** can also be a CMOS-based switch. That is, it can include an NFET **118** coupled in parallel to a PFET **120**, with the drain and source of the NFET **118** tied to the drain and source, respectively, of the PFET **120**. A second enable signal $En2$ is provided to the gate of the NFET **118** and an inverse of the second enable signal $En2b$ is provided to the gate of the PFET **120**. In operation, when the second enable signal $En2$ is high and the inverse of the second enable signal $En2b$ is low, the programming switch **106** is "ON" and the voltage V_{LS} at the input of the programming switch **106** is passed to the output (node n) of the programming switch **106**. When the second enable signal $En2$ is low and the inverse of the second enable signal $En2b$ is high, the programming switch **106** is "OFF" and the voltage V_{LS} at the input of the programming switch **106** is not passed to node n . Alternatively, the programming switch **106** could be any other type of switch suitable for selectively passing V_{LS} to node n when "ON" or preventing passage of V_{LS} to node n when "OFF".

As described above, the programmable transistor **108** has a threshold voltage (VT) that is programmable (e.g., either low or high) in order to selectively adjust the total output current I_{out} of the current mirror circuit **100**. As shown in FIG. 2, the programmable transistor **108** is coupled in parallel with the output FET **112** of the current mirror **102**, with the drain of the programmable transistor **108** connected to the drain of the output FET **112** and the source of the programmable transistor **108** and the source of the output FET **112** tied to ground (VSS). To this extent, the total current I_{out} produced by the current mirror circuit **100** is equal to the sum of the current flowing through the output FET **112** of the current mirror **102** (I_{out_A}) and the current (if any) flowing through the programmable transistor **108** (I_{out_B}).

According to embodiments of the disclosure, the VT of the programmable transistor **108** may be programmed so that it is either low or high. In general, when the VT of the programmable transistor **108** is low, then, during the operation of the current mirror circuit **100**, the total current I_{out} produced by the current mirror circuit **100** is equal to the sum of the current flowing through the output FET **112** of the current mirror **102** (I_{out_A}) and the current flowing

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through the programmable transistor **108** (I_{out_B}). When the VT of the programmable transistor **108** is high, the programmable transistor **108** is off and the total current I_{out} produced by the current mirror circuit **100** is equal to the current flowing through the output FET **112** of the current mirror **102** (I_{out_A}).

According to embodiments of the disclosure, the programmable transistor **108** can be, for example, an NFET (as illustrated) or a PFET configured for electric field-induced VT switching (as opposed to current-induced VT switching) between a first VT (e.g., a low VT) and a second VT (e.g., a high VT) that is different from the first VT and, particularly, that is higher than the first VT. For example, if the programmable transistor **108** is an NFET configured for electric field-induced VT switching, then the VT of the NFET can be programmable to a relatively low VT, such as a VT of 0V or some other low VT, or to a relatively high VT, such as a VT of 1.5V or some VT that is higher than the low VT. FETs that are configured for electric field-induced VT switching and that could be incorporated into the current mirror circuit **100** include, but are not limited to, ferroelectric field effect transistors (FeFETs) and charge trap field effect transistors (CTFETs).

FIGS. 3A and 3B are cross-section diagrams illustrating different VT states of an N-type FeFET **130** that can be incorporated into the current mirror circuit **100** as a programmable transistor **108** according to embodiments of the disclosure. The FeFET **130** can include a semiconductor layer **132** (e.g., a silicon layer or a layer of some other suitable semiconductor material) and, within and/or on the semiconductor layer **132**, N+ source/drain terminals **134** and a channel region **136** (e.g., an intrinsic or P- channel region, depending upon the process technology at issue) positioned between the N+ source/drain terminals **132**. The FeFET **130** can further include a gate terminal **138** adjacent to the channel region **136**. The gate terminal **138** can be a multi-layered structure including, for example, a gate dielectric layer **140** on the channel region **134**, a ferroelectric layer **142** (e.g., a hafnium oxide layer or some other suitable ferroelectric layer) on the gate dielectric layer **140**, and a control gate layer **144** (e.g., a metal gate layer) on the ferroelectric layer **142**. With such a gate terminal **138**, the FeFET **130** has an electric field-based programmable VT and, particularly, has a VT that can be set at a first VT (i.e., a low VT, such as a VT of 0V) or a second VT that is higher than the first VT (i.e., a high VT, such as a VT of 1.5V) depending upon the bias conditions on the gate terminal **138** and at least one of the source/drain terminals **134**. Those skilled in the art will recognize that setting the VT at the first VT is a process often referred to as a writing or programming process, whereas setting the VT at the second VT that is higher than the first VT is often referred to as an erasing process.

To achieve electric field-induced switching to the first VT, the FeFET **130** can be operated in a first programming mode (also referred to herein as a low VT programming mode), during which a programming voltage (VP) that is, for example, within the range of approximately 2.5V to approximately 3.0V can be applied to the gate terminal **138** and a ground voltage (GND) of, for example, 0V can be applied to at least one of the N+ source/drain terminals **134**. This results in the direction of polarization vector of the ferroelectric layer **142** pointing toward the channel region **136** (i.e., it results in +poles of dipoles in the ferroelectric layer **142** being adjacent to the channel region **136** and -poles of the dipoles being adjacent to the control gate layer **144**) such that electrons are attracted to the channel region **136**, thereby setting the VT at the first VT (see FIG. 3A). When the first

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VT is set, the channel region **136** between the N+ source/drain terminals **134** will become conductive in response to a voltage subsequently applied to the gate terminal **138** and set between the first VT and the second VT (and significantly lower than VP).

To achieve electric field-induced switching to the second VT, the FeFET **130** can be operated in a second programming mode (also referred to herein as a high VT programming mode or an erase mode). In the second programming mode, a negative erase voltage V_e can be applied to the gate terminal **138** and GND can be applied to at least one of the N+ source/drain terminals **136**. Alternatively, GND can be applied to the gate terminal **138** and V_e that is, for example, within the range of approximately 2.5V to approximately 3.0V can be applied to at least one of the N+ source/drain terminals **136**. Either way, this results in the direction of polarization vector of the ferroelectric layer **142** pointing toward the control gate layer **144** (i.e., it results in +poles of dipoles in the ferroelectric layer **142** being adjacent to the control gate layer **144** and -poles of the dipoles being adjacent to the channel region **136**) such that electrons are repelled from the channel region **136**, thereby setting the VT to the second VT (see FIG. 3B). When the second VT is set, the channel region **136** will remain non-conductive in response to the same gate voltage (as mentioned above) subsequently applied to the gate terminal **138**.

FIGS. 4A and 4B are cross-section diagrams illustrating different VT states of an N-type CTFET **150** that could, alternatively, be incorporated into the current mirror circuit **100** as a programmable transistor **108**. The N-type CTFET **150** can include a semiconductor layer **152** (e.g., a silicon layer or a layer of some other suitable semiconductor material) and, within and/or on the semiconductor layer **152**, N+ source/drain terminals **154** and a channel region **156** (e.g., an intrinsic or P- channel region, depending upon the process technology at issue) positioned between the N+ source/drain terminals **154**. The N-type CTFET **150** can further include a gate terminal **158** adjacent to the channel region **156**. The gate terminal **158** can be a multi-layered structure including, for example, a gate dielectric layer **160** on the channel region **156**, a charge trap layer **162** (e.g., a silicon nitride layer or some other suitable charge trap layer) on the gate dielectric layer **160**, another gate dielectric layer **164** on the charge trap layer **162** and a control gate layer **166** (e.g., a metal gate layer) on the gate dielectric layer **164**. With such a gate terminal **158**, the N-type CTFET **150** has an electric field-based programmable VT and, particularly, a VT that can be set at a first VT (i.e., a low VT, such as a VT of 0V) or a second VT that is higher than the first VT (i.e., a high VT, such as a VT of 1.5V) depending upon the bias conditions on the gate and source/drain terminals.

To achieve electric field-induced switching to the first VT, the N-type CTFET **150** can be operated in a first program mode (also referred to as a low VT program mode). In the first program mode, a negative voltage ($-VP$) can be applied to the gate terminal **158** and VP can be applied to at least one of the N+ source/drain terminals **154**. This results in electrons moving out of the charge trap layer **162**, thereby setting the VT at the first VT (see FIG. 4A). When the first VT is set, the channel region **156** between the N+ source/drain terminals **154** becomes conductive in response to a voltage subsequently applied to the gate terminal **154** and set between the first VT and the second VT.

To achieve electric field-induced switching to the second VT, the N-type CTFET **150** can be operated in a second program mode (also referred to as a high VT program mode or an erase mode). In the second program mode, V_e can be

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applied to the gate terminal **158** and a negative voltage ($-V_e$) can be applied to at least one of the N+ source/drain terminals **154**. This results in electrons moving into and being trapped by the charge trap layer **162**, thereby setting the VT up at the second VT (see FIG. 4B). When the second VT is set, the channel region **156** between the N+ source/drain terminals **154** will remain non-conductive in response to the same voltage subsequently applied to the gate terminal **158**. It should be noted that the absolute value of V_e can be the same as the absolute value VP or at a different level, but in any case, the voltage should be significantly lower than VP and V_e to avoid unintended VT switching.

FIG. 5 is a table illustrating signals for controlling the operation of the current mirror circuit **100** of FIG. 2 with the programmable FET **108** implemented using a FeFET-type programmable transistor according to embodiments of the disclosure. As shown in the table, the programmable transistor **108** in the current mirror circuit **100** may be operated in a plurality of different modes including an off mode, a current mirror mode, a programming mode, and an erase mode.

The current mirror circuit **100** with the programmable transistor **108** in the off mode is depicted in FIG. 6. When the programmable transistor **108** is in the off mode, the programmable transistor **108** is disabled and does not provide any current to the current mirror circuit **100** ($I_{out_B}=0$). The programmable transistor **108** may be placed in the off mode by turning off the enable switch **104** and turning on the programming switch **106**. The enable switch **104** may be turned off by applying a voltage of $En1=0$ V to the gate of the NFET **114** and applying a voltage of $En1b=VDD$ to the gate of the PFET **116**. The programming switch **106** may be turned on by applying a voltage $En2=VDD$ to the gate of the NFET **118** and applying a voltage $En2b=0$ V to the gate of the PFET **120**. With the programming switch **106** turned on, a voltage $VLS=0$ V is passed from an input of the programmable switch **106** to node n and is applied to the gate of the programmable transistor **108**. This turns off the programmable transistor **108** ($I_{out_B}=0$ V) such that the output current I_{out} of the current mirror circuit **100** is equal to the output current of the current mirror **102** ($I_{out}=I_{out_A}$).

The current mirror circuit **100** with the programmable transistor **108** in the current mirror mode is depicted in FIG. 7. In the current mirror mode, assuming the programmable transistor **108** has been programmed to have a low VT, the total current I_{out} produced by the current mirror circuit **100** is equal to the sum of the current flowing through the output FET **112** of the current mirror **102** (I_{out_A}) and the current flowing through the programmable transistor **108** (I_{out_B}).

The programmable transistor **108** may be placed in the current mirror mode by turning on the enable switch **104** and turning off the programming switch **106**. The enable switch **104** may be turned on by applying a voltage of $En1=VDD$ to the gate of the NFET **114** and applying a voltage of $En1b=0$ V to the gate of the PFET **116**. As such, the voltage V_g at the input of the enable switch **104** is applied to the gate of the programmable transistor **108**. The programming switch **106** may be turned off by applying a voltage $En2=0$ V to the gate of the NFET **118** and applying a voltage $En2b=VDD$ to the gate of the PFET **120**.

With the programmable transistor **108** programmed to have a low VT and with a voltage V_g applied to the gate of the programmable transistor **108**, a current (I_{out_B}) flows through the programmable transistor **108**. To this extent, the output current I_{out} of the current mirror circuit **100** is the sum of the current provided by the current mirror **102**

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(I_{out_A}) and the current flowing through the programmable transistor **108** (I_{out_B}).

The effect of VT on the operation of the current mirror circuit **100** when in the current mirror mode can be appreciated by comparing FIGS. 7 and 8. As detailed above, in the current mirror mode, the enable switch **104** is on (e.g., $En1=VDD$, $En1b=0$ V) and the programming switch **106** is off (e.g., $En2=0$ V, $En2b=VDD$). When the programmable transistor **108** is programmed to have a high VT, the programmable transistor **108** is non-conducting ($I_{out_B}=0$ V). As such, the total current I_{out} produced by the current mirror circuit **100** is equal to the current flowing through the output FET **112** of the current mirror **102** (I_{out_A}).

According to embodiments of the disclosure, the current mirror circuit **100** may include a plurality of programmable transistors **108** connected in parallel to the current mirror **102**. For example, the current mirror circuit **100** depicted in FIG. 9 includes a plurality of programmable transistors **1081**, **1082**, . . . **108_N**. To this extent, the total output current I_{out} of the current mirror circuit **100** is equal to the sum of the current flowing through the output FET **112** (I_{out_A}) and the current flowing through each of the programmable transistors **1081**, **1082**, . . . **108_N** ($I_{out_B1}+I_{out_B2}+. . . I_{out_BN}$).

The current mirror circuit **100** with the programmable transistor **108** in the programming mode is depicted in FIG. 10. During programming mode operations, a programming voltage VP applied to the input of the programming switch may be used to program the programmable transistor **108** to have a low VT.

The programmable transistor **108** may be placed in the programming mode by turning off the enable switch **104** and turning on the programming switch **106**, while setting the voltage $VLS=VP$. The enable switch **104** may be turned off by applying a voltage $En1=0$ V to the gate of the NFET **114** and applying a voltage $En1b=VP$ to the gate of the PFET **116**. The programming switch **106** may be turned on by applying a voltage $En2=VP$ to the gate of the NFET **118** and applying a voltage $En2b=0$ V to the gate of the PFET **120**. To this extent, the voltage at node n and applied to the gate of the programmable transistor **108** is equal to VP.

According to embodiments of the disclosure, VP is variable, which allows the programmable transistor **108** to be programmed to different VTs. In this way, the output current (I_{out_B}) of the programmable transistor **108** when in the current mirror mode may be adjusted to provide different output currents (I_{out_B}). That is, the total current I_{out} of the current mirror circuit **100** ($I_{out}=I_{out_A}+I_{out_B}$) when in the current mirror mode may be adjusted by varying the VT of the programmable transistor **108** based on the value of VP applied in the programming mode.

FIG. 11 depicts the current mirror circuit **100** with the programmable transistor **108** in an erase mode according to embodiments of the disclosure. During erase mode operations, the programmable transistor **108** is erased using an erase voltage ($-Ve$).

The programmable transistor **108** may be placed in the erase mode by turning off the enable switch **104** and turning on the programming switch **106**, while setting the voltage VLS at the programmable switch **106** to the erase voltage $-Ve$. The enable switch **104** may be turned off by applying a voltage $En1=VDD$ to the gate of the PFET **116** and applying a negative voltage to the gate of the NFET **114**. For example, the negative voltage applied to the gate of the NFET **114** may be equal to the sum of the erase voltage $-Ve$ and a negative of the threshold voltage VTn of the NFET **114** (e.g., $En1=-Ve-VTn$). The programming switch **106** may be

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turned on by applying a voltage of $En2=0$ V to the gate of the NFET **118** and applying the erase voltage $En2b=-Ve$ to the gate of the PFET **120**. The voltage at node n and applied to the gate of the programmable transistor **108** is therefore $-Ve$. This places the programmable transistor **108** in a non-conducting state (high VT, $I_{out_B}=0$).

It should be understood that in the structures and method described above, a semiconductor material refers to a material whose conducting properties can be altered by doping with an impurity. Illustrative semiconductor materials include, for example, silicon-based semiconductor materials (e.g., silicon, silicon germanium, silicon germanium carbide, silicon carbide, etc.) and III-V compound semiconductors (i.e., compounds obtained by combining group III elements, such as aluminum (Al), gallium (Ga), or indium (In), with group V elements, such as nitrogen (N), phosphorus (P), arsenic (As) or antimony (Sb)) (e.g., GaN, InP, GaAs, or GaP). A pure semiconductor material and, more particularly, a semiconductor material that is not doped with an impurity for the purposes of increasing conductivity (i.e., an undoped semiconductor material) is referred to in the art as an intrinsic semiconductor. A semiconductor material that is doped with an impurity for the purposes of increasing conductivity (i.e., a doped semiconductor material) is referred to in the art as an extrinsic semiconductor and will be more conductive than an intrinsic semiconductor made of the same base material. That is, extrinsic silicon will be more conductive than intrinsic silicon; extrinsic silicon germanium will be more conductive than intrinsic silicon germanium; and so on. Furthermore, it should be understood that different impurities (i.e., different dopants) can be used to achieve different conductivity types (e.g., P-type conductivity and N-type conductivity) and that the dopants may vary depending upon the different semiconductor materials used. For example, a silicon-based semiconductor material (e.g., silicon, silicon germanium, etc.) is typically doped with a Group III dopant, such as boron (B) or indium (In), to achieve P-type conductivity, whereas a silicon-based semiconductor material is typically doped a Group V dopant, such as arsenic (As), phosphorus (P) or antimony (Sb), to achieve N-type conductivity. A gallium nitride (GaN)-based semiconductor material is typically doped with magnesium (Mg) to achieve P-type conductivity and with silicon (Si) or oxygen to achieve N-type conductivity. Those skilled in the art will also recognize that different conductivity levels will depend upon the relative concentration levels of the dopant(s) in a given semiconductor region. Furthermore, when a semiconductor region or layer is described as being at a higher conductivity level than another semiconductor region or layer, it is more conductive (less resistive) than the other semiconductor region or layer; whereas, when a semiconductor region or layer is described as being at a lower conductivity level than another semiconductor region or layer, it is less conductive (more resistive) than that other semiconductor region or layer.

The method and structure as described above is used in the fabrication of integrated circuit chips. The resulting integrated circuit chips can be distributed by the fabricator in raw wafer form (that is, as a single wafer that has multiple unpackaged chips), as a bare die, or in a packaged form. In the latter case the chip is mounted in a single chip package (such as a plastic carrier, with leads that are affixed to a motherboard or other higher-level carrier) or in a multichip package (such as a ceramic carrier that has either or both surface interconnections or buried interconnections). In any case the chip is then integrated with other chips, discrete circuit elements, and/or other signal processing devices as

part of either (a) an intermediate product, such as a motherboard, or (b) an end product. The end product can be any product that includes integrated circuit chips, ranging from toys and other low-end applications to advanced computer products having a display, a keyboard or other input device, and a center processor.

The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the disclosure. As used herein, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof. “Optional” or “optionally” means that the subsequently described event or circumstance may or may not occur, and that the description includes instances where the event occurs and instances where it does not.

Approximating language, as used herein throughout the specification and claims, may be applied to modify any quantitative representation that could permissibly vary without resulting in a change in the basic function to which it is related. Accordingly, a value modified by a term or terms, such as “about,” “approximately,” and “substantially,” are not to be limited to the precise value specified. In at least some instances, the approximating language may correspond to the precision of an instrument for measuring the value. Here and throughout the specification and claims, range limitations may be combined and/or interchanged, such ranges are identified and include all the sub-ranges contained therein unless context or language indicates otherwise. “Approximately” as applied to a particular value of a range applies to both values, and unless otherwise dependent on the precision of the instrument measuring the value, may indicate $\pm 10\%$ of the stated value(s).

The corresponding structures, materials, acts, and equivalents of all means or step plus function elements in the claims below are intended to include any structure, material, or act for performing the function in combination with other claimed elements as specifically claimed. The description of the present disclosure has been presented for purposes of illustration and description but is not intended to be exhaustive or limited to the disclosure in the form disclosed. Many modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope and spirit of the disclosure. The embodiment was chosen and described in order to best explain the principles of the disclosure and the practical application, and to enable others of ordinary skill in the art to understand the disclosure for various embodiments with various modifications as are suited to the particular use contemplated.

The invention claimed is:

1. A structure, comprising:

a current mirror;

a programmable transistor with an adjustable threshold voltage connected in parallel with an output transistor of the current mirror;

an enable switch for coupling a gate of the output transistor to a gate of the programmable transistor; and

a programming switch coupled to the gate of the programmable transistor,

wherein the gate of the output transistor of the current mirror is coupled to an input of the enable switch and

an output of the enable switch is coupled to the gate of the programmable transistor, and

wherein a drain of the programmable transistor is connected to a drain of the output transistor of the current mirror, and wherein a source of the programmable transistor and a source of the output transistor of the current mirror are coupled to ground.

2. The structure according to claim 1, wherein the programmable transistor comprises any of a ferroelectric field effect transistor and a charge trap field effect transistor.

3. The structure according to claim 1, wherein each of the enable switch and the programming switch comprises an N-type field effect transistor (FET) connected in parallel with a P-type FET.

4. The structure according to claim 1, wherein the adjustable threshold voltage of the programmable transistor is programmable between a first threshold voltage and a second, higher threshold voltage.

5. The structure according to claim 4, wherein the programmable transistor is on when programmed at the first threshold voltage and when a gate voltage on the output transistor of the current mirror is passed to the gate of the programmable transistor by the enable switch, and wherein a total current output by the structure is equal to a sum of an output current of the current mirror and a current flowing through the programmable transistor.

6. The structure according to claim 5, wherein the programmable transistor is off when programmed at the second threshold voltage, and wherein the total current output by the structure is equal to the output current of the current mirror.

7. The structure according to claim 4, wherein, to program the programmable transistor, the enable switch is off and the programming switch is on, and wherein a programming voltage applied to an input of the programming switch is applied to the gate of the programmable transistor.

8. The structure according to claim 7, wherein the programming voltage is adjustable, and wherein the programmable transistor is programmable at a plurality of different first threshold voltages based on the adjustable programming voltage.

9. The structure according to claim 4, wherein, to erase the programmable transistor, the enable switch is off and the programming switch is on, and wherein an erase voltage applied to an input of the programming switch is applied to the gate of the programmable transistor.

10. The structure according to claim 4, wherein, in a current mirror mode, the enable switch is on and the programming switch is off, and wherein a voltage applied to the gate of the output transistor of the current mirror is applied to the gate of the programmable transistor.

11. The structure according to claim 1, further comprising a plurality of the programmable transistors connected in parallel with the current mirror.

12. A current mirror circuit, comprising:

a current mirror;

a programmable transistor with an adjustable threshold voltage connected in parallel with an output transistor of the current mirror, wherein the programmable transistor is operable in any of an off mode, a current mirror mode, a programming mode, and an erase mode;

an enable switch for coupling a gate of the output transistor to a gate of the programmable transistor; and

a programming switch coupled to the gate of the programmable transistor for controlling the adjustable threshold voltage of the programmable transistor,

wherein the gate of the output transistor of the current mirror is coupled to an input of the enable switch and

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an output of the enable switch is coupled to the gate of the programmable transistor, and wherein a drain of the programmable transistor is connected to a drain of the output transistor of the current mirror, and wherein a source of the programmable transistor and a source of the output transistor of the current mirror are coupled to ground.

13. The current mirror circuit according to claim 12, wherein the programmable transistor comprises any of a ferroelectric field effect transistor and a charge trap field effect transistor.

14. The current mirror circuit according to claim 13, wherein the adjustable threshold voltage of the programmable transistor is programmable between a first threshold voltage and a second, higher threshold voltage.

15. The current mirror circuit according to claim 14, wherein:

the programmable transistor is on when programmed at the first threshold voltage and when in the current mirror mode, wherein a total current output by the current mirror structure is equal to a sum of an output current of the current mirror and a current flowing through the programmable transistor; and

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the programmable transistor is off when programmed at the second threshold voltage and when in the off mode, wherein the total current output by the current mirror structure is equal to the output current of the current mirror.

16. The current mirror circuit according to claim 14, wherein, to program the programmable transistor in the programming mode, the enable switch is off and the programming switch is on, wherein an adjustable programming voltage applied to an input of the programming switch is applied to the gate of the programmable transistor, and wherein the programmable transistor is programmable at a plurality of different first threshold voltages based on the adjustable programming voltage.

17. The current mirror circuit according to claim 14, wherein, to erase the programmable transistor in the erase mode, the enable switch is off and the programming switch is on, and wherein an erase voltage applied to an input of the programming switch is applied to the gate of the programmable transistor.

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